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Tattoo ink mixing apparatus

Abstract

A tattoo ink mixing apparatus has a housing having a cavity, a plunger movable between a first position and a second position, and a dynamic member disposed in the cavity and adapted to engage with the plunger distal end. The dynamic member moves in response to movement of the plunger and alternatively creates a suction force and an expulsion force in the opening of the distal end of the housing as the plunger moves between its first and second positions.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 16/782,701, filed Feb. 5, 2020 and issued as U.S. Pat. No. 11,400,425 on Aug. 2, 2022, which claims the benefit of U.S. Provisional Application No. 62/801,233, filed Feb. 5, 2019. The entire disclosure of each of these related applications is hereby incorporated by reference into this disclosure.

FIELD

(1) The disclosure relates generally to the field of tattoos. More specifically, the disclosure relates to tattoo ink mixing apparatuses and tattoo machine systems useful for mixing tattoo ink. Particular embodiments relate to tattoo ink mixing cartridges that utilize a tattoo machine's axial motion to mix tattoo ink.

BACKGROUND

(2) A need exists for tattoo ink mixing apparatuses, tattoo ink mixing cartridges to be used with tattoo machines, and tattoo machine systems.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of an example tattoo ink mixing apparatus according to one embodiment.
- (2) FIG. 2 is a perspective view of another example tattoo ink mixing apparatus with a dynamic member in a starting position.
- (3) FIG. 3 is a perspective view of the tattoo ink mixing of FIG. 2 with the dynamic member in a first configuration.
- (4) FIG. 4 is a perspective view of the tattoo ink mixing of FIG. 2 with the dynamic member in a second configuration.
- (5) FIG. 5 is a perspective view of another example tattoo ink mixing apparatus with a dynamic member in a starting position.
- (6) FIG. 6 is a perspective view of the tattoo ink mixing of FIG. 5 with the dynamic member in a first configuration.
- (7) FIG. 7 is a perspective view of the tattoo ink mixing of FIG. 5 with the dynamic member in a second configuration.
- (8) FIG. 8 is a perspective view of another example tattoo ink mixing apparatus.
- (9) FIG. 9 is a perspective view of an example tattoo machine system including the tattoo ink mixing apparatus illustrated in FIG. 8.

DETAILED DESCRIPTION

- (10) The following detailed description and the appended drawings describe and illustrate various example embodiments of tattoo ink mixing apparatuses and tattoo machine system. The description and illustration of these examples are provided to enable one skilled in the art to make and use a tattoo ink mixing apparatus. They are not intended to limit the scope of the claims in any manner.
- (11) FIG. 1 illustrates an example embodiment of a tattoo ink mixing apparatus **10**. The tattoo ink mixing apparatus **10** comprises a cartridge **12** having a lengthwise axis **13**, a cartridge housing **14**, a plunger **16**, a dynamic member **18**, and a cartridge distal end **20** defining an opening **22**.
- (12) The cartridge housing **14** comprises a housing proximal end **24**, a housing distal end **26**, and a housing body **28** extending from the housing proximal end **24** to the housing distal end **26**. The housing proximal end **24** defines a housing proximal end opening **30** that provides access to a cavity **36** located within the housing body **28**. The housing body **28** defines an inside wall **38**, a housing proximal portion **32**, an intermediate portion **33**, and a housing distal portion **34**. The housing proximal portion **32** extends from the housing proximal end **24** to the housing intermediate portion **33** and has a first length **40**. The housing intermediate portion **33** tapers from the housing proximal portion **32** to the housing distal portion **34** and has a second length **44**. The housing distal portion **34** extends from the housing intermediate portion **33** to the housing distal end **26** and has a third length **46**. In the illustrated embodiment, the first length **40** is greater than each of the second length **44** and the third length **46**.
- (13) The housing proximal portion **32** has a first diameter **42** and the housing distal portion **34** has a second diameter **48**. In the illustrated embodiment, the first diameter **42** is greater than the second diameter **48**.
- (14) While the housing body **28** has been illustrated as having an intermediate portion that tapers from the proximal portion to the distal portion, the tapered intermediate portion can be omitted and the housing body can comprise a single portion. Alternatively, the housing body can have multiple tapered intermediate portions and skilled artisans will be able to select an appropriate number of tapered intermediate portions based on various considerations. Example number of tapered intermediate portions that are considered suitable include one, more than one, a plurality, two, three, or any other number considered suitable for a particular embodiment.
- (15) While the housing proximal portion **32** has been illustrated as having a first length **40** that is greater than each of the second length **44** of housing intermediate portion **33** and the third length **46**

of housing distal portion **34**, and a first diameter **42** that is greater than the second diameter **48** of housing distal portion **34**, any suitable lengths and diameters can be used and a skilled artisan will be able to select an appropriate first length, second length, third length, first diameter, and second diameter based on various considerations. Example lengths considered suitable for a first, second, and third lengths include a first length that is equal to, substantially equal to, or less than each of the second length and the third length. Example diameters considered suitable for a first and second diameters include a first diameter that is equal to, substantially equal to, greater than, or less than the second diameter.

(16) The housing body **28** defines the cavity **36** that has a cavity mixing chamber **104** and a cavity proximal chamber **106**. The cavity mixing chamber **104** extends from a first location **108** between the housing proximal end **24** and the housing distal end **26** to the opening **22** defined on the cartridge distal end **20**. The opening **22** provides direct access to the cavity mixing chamber **104**. The cavity proximal chamber **106** extends from the opening **30** defined on the housing proximal end **24** to the first location **108**. The opening **30** provides direct access to the cavity proximal chamber **106**. The cavity proximal chamber **106** is adapted to receive the distal end **62** and the second portion **68** of the plunger **16** such that the distal end **62** and the second portion **68** of the plunger **16** are disposed in the cavity proximal chamber **106**.

(17) Opening **30** has a diameter **31** that is equal to the first diameter **42** of housing proximal portion **32**. Each of openings **22** and **30** provides direct access to the cavity **36** such that any material disposed through either of openings **22** and **30** is disposed in the cavity **36**. Each of opening **30** and opening **22** can have any suitable shape and size and skilled artisans will be able to select an appropriate shape and size for the openings based on various considerations. Example shapes considered suitable for the openings include round, oval, and any other shapes considered suitable for a particular embodiment.

(18) Plunger **16** comprises a proximal end **60**, a distal end **62**, an elongate body **64**, a first portion **66**, and a second portion **68**. Elongate body **64** extends from the proximal end **60** to the distal end **62** and has a plunger diameter **70**. In the illustrated embodiment, plunger diameter **70** is substantially less than the first diameter of the proximal portion **40** of the housing body **28**. Skilled artisans will be able to select an appropriate plunger diameter based on various considerations.

(19) The plunger first portion **66** extends from the plunger proximal end **60** to a center point **72** located between the plunger proximal end **60** and the plunger distal end **62** and has a first length **74** measured along the lengthwise axis **13**. The plunger second portion **68** extends from the center point **72** to the plunger distal end **62** and has a second length **76** measured along the lengthwise axis **13**. In the illustrated embodiment, the center point **72** lies halfway between the proximal end **60** and the distal end **62**, measured along the lengthwise axis **13**, such that the first length **74** is equal to the second length **76**. Alternatively, the center point **72** can lie anywhere between the proximal end **60** and the distal end **62**. For example, the center point **72** can lie between the proximal end **60** and the distal end **62**, measured along the lengthwise axis **13**, such that the first length is substantially equal to the second length, the first length is greater than the second length, or the first length is less than the second length. Skilled artisans will be able to select an appropriate location for the center point **72** based on various considerations.

(20) The distal end **62** of the plunger **16** is disposed through the opening **30** defined by the housing proximal end **24** such that the plunger second portion **68** is fully disposed in the cavity **36**, the center point **72** lies adjacent the housing proximal end **24**, and the plunger first portion **66** is fully disposed outside of the cavity **36**. This structural arrangement provides a starting position for the plunger **16** in which the center point **72** is disposed a first distance **84** from opening **22**, measured along the lengthwise axis **113**.

(21) While plunger **16** has been described as being disposed through opening **30** such that the plunger second portion **68** is disposed completely in cavity **36**, center point **72** lies on housing proximal end **24**, and plunger first portion **66** is disposed completely outside of cavity **36**, any

portion of the plunger **16** can be disposed through the opening **30** and in the cavity **36** and skilled artisans will be able to select an appropriate portion for the plunger **16** to be disposed through the opening **30** based on various considerations. Example portions of the plunger considered suitable to be disposed through an opening include the plunger first portion **66**, the plunger second portion **68**, a portion of the plunger first portion **66**, a portion of the second portion **68**, both the first portion **66** and the second portion **68**, and neither of the first portion **66** or the second portion **68**.

(22) In the illustrated embodiment, the diameter **70** of the plunger **16** is less than the first diameter **42** of housing proximal portion **32**. Alternatively, plunger **16** can have a diameter that is equal to, substantially equal to, or substantially less than the diameter of housing proximal portion **32** and skilled artisans will be able to select an appropriate plunger diameter based on various considerations.

(23) While the plunger **16** has been illustrated as defining a first portion **66** having a first length **74** that is substantially equal to the second length **76** of the second portion **68**, the plunger **16** can define any suitable first and second lengths and skilled artisans will be able to select appropriate first and second lengths based on various considerations. Example lengths that are considered suitable include a first length that is equal to, greater than, or less than the second length.

(24) The plunger **16** is adapted to move axially along the lengthwise axis **13** between the starting position, a first position, and a second position. To move the plunger **16** between the starting position and the first position distal force is applied along the lengthwise axis **13** to the plunger proximal end **60**. When the plunger **16** is in the first position the plunger center point **72** is disposed inside the cavity **36** distal to the housing proximal end opening **30** and a portion of the plunger first portion **66** is disposed inside the cavity **36**. In this position, the center point **72** of the plunger **16** lies a second distance **86** along the lengthwise axis **13** from the cartridge distal end opening **22**. The second distance **86** is less than the first distance **84**. To move the plunger **16** from the first position to the starting position, proximal force is applied to the plunger proximal end **60** along the lengthwise axis **13** such that the plunger center point **72** is disposed adjacent the opening **30**. To move the plunger from the starting position to the second position, proximal force is applied to the plunger proximal end **24** along the lengthwise axis **13**. In the second position, the plunger center point **72** is disposed outside the cavity **36** proximal to housing proximal end opening **30** and a portion of the plunger second portion **68** is disposed outside the cavity **36**. In this position, the center point **72** lies a third distance **88** along the lengthwise axis **13** from the cartridge distal end opening **22**. The third distance **88** is greater than each of the first distance **84** and the second distance **86**. To move the plunger **16** from the second position to the starting position **84**, distal force is applied to the plunger proximal end **24** along the lengthwise axis **13** until the plunger center point **72** moves from being disposed outside the cavity **36** to being disposed adjacent the opening **30**.

(25) In practice, when the plunger **16** is in the first position, the center point is disposed outside of the opening **22** defined on the cartridge distal end **20**. As distal force is applied to the plunger proximal end **60** along the lengthwise axis **13**, the plunger **16** moves from the first position towards the second position. The center point **72** that is disposed outside the cavity **36** moves distally along the lengthwise axis **13** toward the opening **30**. When the plunger **16** is advanced to the starting position, the center point **72** lies adjacent the opening **30**. As distal force continues to be applied to the plunger proximal end **60** along the lengthwise axis **13**, the plunger **16** advances from the starting position to the second position and the center point **72** moves distally along the lengthwise axis **13** until it is disposed in the cavity **36**.

(26) As proximal force is applied to the plunger proximal end **60** along the lengthwise axis **13**, the plunger **16** moves from the second position toward the first position. The center point **72** is disposed in the cavity **36** and moves proximally along the lengthwise axis **13** toward the opening **30** defined on the housing proximal end **24**. When the plunger **16** is advanced to the starting position, the center point **72** lies adjacent the opening **30**. As proximal force continues to be applied

to the plunger proximal end **60** along the lengthwise axis **13**, the plunger **16** advances from the starting position to the first position and the center point **72** moves proximally along the lengthwise axis **13** until it is disposed outside the cavity **36**.

(27) The dynamic member **18** defines a center portion **90**, a center portion diameter **92**, an outer circumference **94**, an outer diameter **96**, a starting configuration, a first configuration and a second configuration.

(28) In the illustrated embodiment, the dynamic member **18** is disposed in the cavity **36** defined by the housing body **28** such that the dynamic member **18** is fully disposed within the proximal portion **32** of the housing body **28**.

(29) The outer diameter **96** of the dynamic member **18** is substantially equal to the diameter **42** of the housing proximal portion **32** such that the outer circumference **94** lies adjacent the inside wall **38** of the housing body **28**. In the illustrated embodiment, the outer circumference **94** is fixedly attached to the inside wall **38** of the housing body **28**. Any suitable type of attachment can be used and skilled artisans will be able to select an appropriate type of attachment based on various considerations, including gluing.

(30) While the tattoo ink mixing apparatus **10** has been illustrated as comprising a cartridge housing **14** and a dynamic member **18** as separate and distinct parts, the tattoo ink mixing apparatus **10** can comprise a cartridge housing **14** with a preformed dynamic member **18**.

(31) While the dynamic member **18** has been illustrated as being disposed in the cavity **36** within the housing proximal portion **32** and as having an outer diameter **96** that is substantially equal to the first diameter **42** of the housing proximal portion **32**, the dynamic member **18** can be disposed anywhere in the cavity **36** and the dynamic member can have any suitable diameter. For example, the dynamic member **18** can be disposed in the cavity **36** within the distal portion **34** such that the dynamic member **18** has an outer diameter **96** that is equal to the second diameter **48** of the housing distal portion **34**, or the dynamic member **18** can be disposed in the cavity **36** at a location within the intermediate portion **33** such that the dynamic member **18** has an outer diameter **96** that is equal to the diameter of the intermediate portion **33** at that location. Skilled artisans will be able to select appropriate locations where the dynamic member outer circumference is attached to the cartridge housing inside wall based on various considerations.

(32) While the outer circumference **94** of the dynamic member **18** has been described as being fixedly attached to the inside wall **38** of the housing body **28**, any suitable type of attachment can be made and skilled artisans will be able to select a suitable type of attachment between the dynamic member outer circumference and the cartridge housing inside wall based on various considerations. Example types of attachments considered suitable include fixed attachment, releasable attachment, and any other attachment considered suitable for a particular embodiment.

(33) The center portion **90** of the dynamic member **18** is adapted to engage with the plunger distal end **62** and has a center portion diameter **92** that is substantially equal to the plunger diameter **70**. Axial movement of the plunger distal end **62** along the lengthwise axis **13** as the plunger **16** moves between the starting position **72**, the first position, and the second position causes a proportional movement in the center portion **90** of the dynamic member **18** and induces the dynamic member **18** to move between the starting configuration, the first configuration, and the second configuration.

(34) The dynamic member **18** is adapted to move from the starting configuration to the first configuration by applying distal force to the plunger proximal end **60** along the lengthwise axis **13**. As distal force is applied, the plunger **16** moves from the starting position, where the plunger center point **72** is disposed adjacent the housing proximal end opening **30**, to the first position, where the plunger center point **72** is disposed in the cavity **36**, and the plunger distal end **62** induces the dynamic member **18** to move from the starting configuration to the first configuration. In the first configuration **100** the plunger center point **72** is disposed in the cavity **36**. To move the dynamic member **18** from the first configuration to the starting configuration, proximal force is applied to the plunger proximal end **60** along the lengthwise axis **13**. As proximal force is applied, the plunger

16 moves from the first position to the starting position and the plunger distal end **62** induces the dynamic member **18** to move from the first configuration to the starting configuration. In the starting configuration, the plunger center point **72** is disposed adjacent the housing proximal end opening **30**. As proximal force continues to be applied to the plunger proximal end **60** along the lengthwise axis **13**, the plunger **16** moves from the starting position to the second position, and the plunger distal end **62** induces the dynamic member **18** to move from the starting configuration to the second configuration. In the second configuration, the plunger center point **72** is disposed proximal to the housing proximal end opening **30** such that the center point **72** is disposed outside the cavity **36**. To move the dynamic member **18** from the second configuration to the starting configuration, distal force is applied to the plunger proximal end **60** along the lengthwise axis **13**. As distal force is applied, the plunger **16** moves from the second position to the starting position and the plunger distal end **62** induces the dynamic member **18** to move from the second configuration to the starting configuration.

(35) In practice, the cartridge distal end **20** is optionally disposed in an ink reservoir containing tattoo ink (not shown) such that the opening **22** and a portion of the housing distal portion **34** are disposed in the ink. When proximal force is applied to the plunger proximal end **60** along the lengthwise axis **13**, the plunger distal end **62** moves proximally along the lengthwise axis **13** and engages with the center portion **90** of the dynamic member **18**. This engagement moves the dynamic member **18** from the starting configuration or the first configuration to the second configuration. This change in configurations causes suction force to occur in the housing cavity **36** and causes ink to be drawn up through the opening **22** and into the cavity **36**. When distal force is applied to the plunger proximal end **60** along the lengthwise axis **13**, the plunger distal end **62** moves distally along the lengthwise axis **13** and engages with the center portion **90** of the dynamic member **18**. This engagement with the center portion **90** moves the dynamic member **18** from the starting configuration or the second configuration to the first configuration. This change in configurations causes an expulsion force to occur in the housing cavity **36** and causes ink to be expelled out of the opening **22**.

(36) FIGS. **2**, **3**, and **4** illustrate another example embodiment of a tattoo ink mixing apparatus **110**. Tattoo ink mixing apparatus **110** is similar to the tattoo ink mixing apparatus **10** illustrated in FIG. **1** and described above, except as detailed below. Reference numbers in FIGS. **2**, **3**, and **4** refer to the same structural element or feature referenced by the same numbers in FIG. **1**, offset by **100**. Thus, the tattoo ink mixing apparatus **110** comprises a cartridge **112** having a lengthwise axis **113**, a cartridge housing **114**, a plunger **116**, a dynamic member **118**, and a cartridge distal end **120** defining an opening **122**.

(37) In this embodiment, the cavity **136** defines an inside wall **138**, a cavity mixing chamber **218**, and a cavity proximal chamber **220**. The inside wall **138** extends from the opening **130** defined by the housing proximal end **124** to the opening **122** defined on the cartridge distal end **120**. The cavity proximal chamber **220** extends from the opening **130** to a first location **222** between the housing proximal end **124** and the housing distal end **126**. The cavity proximal chamber **220** has a first diameter **221** and is adapted to receive the second portion **168** of the plunger **116** such that the plunger center point **172** lies adjacent the opening **130**.

(38) The cavity mixing chamber **218** extends from the first location **222** to the opening **122** defined on the cartridge distal end **120**. The cavity mixing chamber **218** comprises a proximal portion **224**, an intermediate portion **226**, and a distal portion **228**. The proximal portion **224** extends from the first location **222** to the intermediate portion **226** and has a proximal portion length **230** and a proximal portion diameter **232**. The intermediate portion **226** tapers from the proximal portion **224** to the distal portion **228** and has an intermediate portion length **234**. The distal portion **228** extends from the intermediate portion **226** to the opening **122** and has a distal portion length **236** and a distal portion diameter **238**. In the illustrated embodiment, the proximal portion diameter **232** is equal to the first diameter **221** of the cavity proximal portion **220** and is greater than the distal

portion diameter **238**.

(39) In this embodiment, the distal portion **228** of the cavity mixing chamber **218** defines a plurality of passageways **250** that individually provide fluid communication between the cavity mixing chamber **218** and an environment external to the cartridge housing **114**. While the illustrated embodiment includes a plurality of passageways **250**, it is noted that only one or two passageways can be included on a housing distal portion according to a particular embodiment. Indeed, any suitable number of passageways can be included. Furthermore, the passageways on a housing distal portion according to a particular embodiment can have any suitable size, shape, configuration, arrangement, and location. The number, size, shape, configuration, arrangement, and location of passageways included on a housing distal portion according to a particular embodiment can be selected by a skilled artisan based on various considerations. The inventor(s) have determined that a plurality of passageways that comprises opposably positioned series of passageways is suitable for a cartridge intended to be used to mix tattoo ink in an ink reservoir.

(40) As illustrated in FIGS. **3** and **4**, and as will be described in more detail herein, the cavity mixing chamber **218** is adapted to receive tattoo ink (not shown). As the dynamic member **118** moves from the second configuration to the first configuration, the internal pressure of the cavity mixing chamber **218** decreases to a level below the pressure of the environment external to the cartridge housing **114**. This causes ink to be drawn through the opening **122** defined on the cartridge distal end **120** and/or one or more passageway **252** of the plurality of passageways **250** and into the distal portion **228** of the cavity mixing chamber **218**. The ink then travels proximally through the distal portion **228** towards the intermediate portion **226** and the proximal portion **224** of the cavity mixing chamber **218**. As the dynamic member **118** continues to move toward the first configuration, the ink travels proximally through the intermediate portion **226** and into the proximal portion **224** of the cavity mixing chamber **218**. When the dynamic member **118** is in the first configuration, the cavity mixing chamber **218** is substantially full of ink. As the dynamic member **118** moves from the first configuration to the second configuration, the internal pressure of the cavity mixing chamber **218** increases to a level greater than the pressure of the environment external to the cartridge housing **114**. This causes ink to travel distally from the proximal portion **224** to the distal portion **228**, passing through the intermediate portion **226**. As the dynamic member **118** continues to move toward the second configuration, the ink is expelled through the opening **122** and/or one or more passageway **252** of the plurality of passageways **250**. When the dynamic member **118** is in the second configuration, the cavity mixing chamber **218** is substantially empty of ink.

(41) As illustrated in FIGS. **3** and **4**, the distal end **162** of the plunger **116** is fixedly attached to the first side **204** of the center portion **190** of the dynamic member **118**. When the plunger **116** is in the starting position, the plunger distal end **162** lies adjacent the first location **222** and the plunger center point **172** lies adjacent the opening **130** defined on the housing proximal end **124**. In the starting position, the plunger distal end **162** is disposed a first distance **184** from the opening **122** defined on the cartridge distal end **120**, measured along the lengthwise axis **113**. When the plunger **116** is in the first position, the plunger distal end **162** lies distal to the first location **222** and is disposed in the proximal portion **224** of the cavity mixing chamber **218**. The plunger distal end **162** is disposed a second distance **186** from the opening **122**, measured along the lengthwise axis **113**. In the first position, the plunger center point **172** lies distal to the opening **130** and is disposed inside the proximal chamber **220** of the cavity **136**. When the plunger **116** is in the second position, the plunger distal end **162** lies proximal to the first location **222** and is disposed in the proximal chamber **220** of the cavity **136**. The plunger distal end **162** is disposed a third distance **188** from the opening **122**, measured along the lengthwise axis **113**. In the second position, the plunger center point **172** lies proximal to the opening **130** and is disposed outside of the proximal chamber **220** of the cavity **136**.

(42) In this embodiment, the dynamic member **118** defines a center portion **190**, an outer

circumference **194**, an outer diameter **196**, a starting configuration, a first configuration and a second configuration.

(43) The center portion **190** of the dynamic member **118** has a first side **204**, a second side **206**, and a center portion diameter **192**. The diameter **192** of the center portion **190** is substantially equal to the diameter **170** of the plunger **116**. The first side **204** is fixedly attached to the distal end **162** of the plunger **116** and the second side **206** is in communication with the proximal portion **224** of the cavity mixing chamber **218**. Any suitable type of attachment can be used and skilled artisans will be able to select an appropriate type of attachment based on various considerations. Example suitable types of attachments include gluing or any other type of attachment considered suitable for a particular embodiment.

(44) While the first side **204** of the center portion **190** of the dynamic member **118** has been described as being fixedly attached to the plunger distal end **162**, any suitable type of attachment can be made and skilled artisans will be able to select a suitable type of attachment between the center portion and the plunger distal end based on various considerations, including releasable attachment and any other attachment considered suitable for a particular embodiment.

(45) The dynamic member **118** is disposed in the cavity **136** such that the dynamic member **118** lies adjacent the first location **222**. In the illustrated embodiment, the dynamic member **118** functions as a sealing member and comprises an elastic membrane **119** that has an outer diameter **196**. In the illustrated embodiment, the outer diameter **196** of the dynamic member **118** is substantially equal to the first diameter **221** of the cavity proximal chamber **220**. The outer circumference **194** of the dynamic member **118** lies adjacent the inside wall **138** of the cavity **136** and is fixedly attached to the inside wall **138** at the first location **222**. Such a structural configuration results in the dynamic member **118** forming a tight seal between the cavity mixing chamber **218** and the cavity proximal chamber **220** such that tattoo ink is not able to travel from the proximal portion **224** of the cavity mixing chamber **218** to the cavity proximal chamber **220**.

(46) As illustrated in FIGS. **3** and **4**, the dynamic member **118** is adapted to move between a starting configuration, a first configuration, and a second configuration. In the starting configuration, the plunger is in the starting position and the center portion **190** of the dynamic member **118** is disposed adjacent the first location **222**. Each of the center portion **190** and the outer circumference **194** extends a first distance **208** from the opening **122** defined on the cartridge distal end **120**, measured along the lengthwise axis **113**. The first distance **208** is equal to the first distance **184** of the distal end **162** of the plunger **116**. When distal force is applied to the plunger proximal end **160** along the lengthwise axis **113**, the plunger distal end **162** pushes the center portion **190** of the dynamic member **118** distally until the center portion **190** is disposed a second distance **210** from the opening **122**, measured along the lengthwise axis **113**. The second distance **210** of the center portion **190** is equal to the second distance **186** of the plunger distal end **162**. In the second configuration, the center portion **190** is disposed distal to the outer circumference **194** of the dynamic member **118** and distal to the first location **222** such that the center portion **190** is disposed in the proximal portion **224** of the cavity mixing chamber **218**. When proximal force is applied to the plunger proximal end **160** along the lengthwise axis **113**, the plunger distal end **162** pulls the center portion **190** of the dynamic member **118** proximally until the center portion **190** is disposed a third distance **212** from the opening **122**, measured along the lengthwise axis **113**. The third distance **212** of the center portion **190** is equal to the third distance **188** of the plunger distal end **162**. In the third configuration, the center portion **190** is disposed proximal to the outer circumference **194** of the dynamic member **118** and proximal to the first location **222** such that the center portion **190** is disposed in the proximal chamber **220** of the cavity **136**.

(47) In practice, the cartridge distal end **120** is optionally disposed in an ink reservoir containing tattoo ink (not shown) such that a portion of the distal portion **228** of the cavity mixing chamber **218**, the opening **22** defined on the cartridge distal end **120**, and at least one passageway **252** of the plurality of passageways **250** are disposed in the ink. When proximal force is applied to the plunger

proximal end **160** along the lengthwise axis **113**, the dynamic member **118** moves from the second configuration to the first configuration. During this transition from the second configuration to the first configuration, the plunger distal end **162** and the center portion **190** of the dynamic member **118** move proximally along the lengthwise axis **113** until the plunger distal end **162** and the center portion **190** are disposed in the proximal chamber **220** of the cavity **136**. As the center portion **190** moves proximally towards the second distance **210**, the internal pressure inside the cavity mixing chamber **218** decreases to a level less than the pressure of the environment external to the cartridge housing **114**. This causes ink to be drawn through the opening **122** defined on the cartridge distal end **120** and/or one or more passageways **252** of the plurality of passageways **250** and into the distal portion **228** of the cavity mixing chamber **218**. The ink then travels proximally through the distal portion **228** and into the intermediate portion **226** and proximal portion **224** of the cavity mixing chamber **218**. When the dynamic member **118** is in the first configuration, the cavity mixing chamber **218** is substantially full of ink.

(48) When distal force is applied to the plunger proximal end **160** along the lengthwise axis **113**, the dynamic member **118** moves from the first configuration to the second configuration. During this transition from the first configuration to the second configuration, the plunger distal end **162** and the center portion **190** of the dynamic member **118** move distally along the lengthwise axis **113** until the plunger distal end **162** and the center portion **190** are disposed in the proximal portion **224** of the cavity mixing chamber **218**. As the center portion **190** moves distally towards the first distance **208**, the internal pressure inside the cavity mixing chamber **218** increases to a level greater than the pressure of the environment external to the cartridge housing **114**. As the dynamic member **118** moves from the second configuration to the first configuration, the internal pressure of the cavity mixing chamber **218** drops below the pressure of the environment external to the cartridge housing **114**. This causes the ink to travel distally through the proximal portion **224**, into the intermediate portion **226**, and into the distal portion **228**. As the dynamic member **118** continues to move toward the second configuration, the ink is expelled through one or more passageways **252** of the plurality of passageways **250** and/or the opening **122** defined on the cartridge distal end **120**. When the dynamic member **118** is in the second configuration, the cavity mixing chamber **218** is substantially empty of ink. The repeated movement of the dynamic member **118** between the first configuration and the second configuration causes ink to be continuously drawn and expelled through the opening **122** and/or one or more passageways **252** of the plurality of passageways **250** causing the ink to be stirred.

(49) While the cartridge **112** has been described as having a dynamic member **118** that forms a tight seal between the cavity mixing chamber **218** and the cavity proximal chamber **220** to prevent ink from traveling from the proximal portion **224** of the cavity mixing chamber **218** to the cavity proximal chamber **220**, any suitable sealant can be used and skilled artisans will be able to select a suitable sealant based on various considerations. For example, FIGS. **5**, **6**, and **7** illustrate another example embodiment of the tattoo ink mixing apparatus **510**. Tattoo ink mixing apparatus **510** is similar to the tattoo ink mixing apparatus **110** illustrated in FIG. **2** and described above, except as detailed below. Reference numbers in FIGS. **5**, **6**, and **7** refer to the same structural element or feature referenced by the same numbers in FIG. **2**, offset by **400**. Thus, the tattoo ink mixing apparatus **510** comprises a cartridge **512** having a lengthwise axis **513**, a cartridge housing **514**, a plunger **516**, a dynamic member **518**, and a cartridge distal end **520** defining an opening **522**.

(50) In this embodiment, the tattoo ink mixing apparatus **510** comprises a cartridge **512** that further defines a sealing membrane **550**. The sealing membrane **550** is disposed distal to the dynamic member **518** in the proximal portion **624** of the cavity mixing chamber **618**. The sealing membrane **550** has a diameter **552** that is substantially equal to each of the outer diameter **596** of the dynamic member **518** and the diameter **632** of the proximal portion **624** of the cavity mixing chamber **618**. The sealing membrane **550** has a circumference **554** that is fixedly attached to the inside wall **538** of the cavity **536** at a location **540** between the first location **622** and the intermediate portion **626**.

The sealing membrane **550** forms a tight seal between a first portion **690** of the proximal portion **624** that is proximal to the sealing membrane **550** and a second portion **692** of the proximal portion **624** that is distal to the sealing membrane **550**. The sealing membrane **550** prevents ink from traveling from the second portion **692** to the first portion **690**. The first portion **690** extends from the dynamic member **518** to the sealing membrane **550** and is sealed proximally by the dynamic member **518**, distally by the sealing membrane **550**, and circumferentially by the cartridge housing **514**.

(51) As illustrated in FIGS. **6** and **7**, the sealing membrane **550** defines a starting configuration, a first configuration, and a second configuration. In the starting configuration, the dynamic member **518** is in the starting configuration and the center **706** of the sealing membrane **550** and the circumference **554** of the sealing member **550** lie on a plane **708** that is orthogonal to the lengthwise axis **513**. In the first configuration, the dynamic member **518** is in the first configuration and the center **706** of the sealing membrane **550** lies distal to the circumference **554** of the sealing member **550** such that the center **706** is disposed in the second portion **692** of the proximal portion **624** of the cavity mixing chamber **618**. In the second configuration, the dynamic member **518** is in the second configuration and the center **706** of the sealing membrane **550** lies proximal to the circumference **554** of the sealing member **550** such that the center **706** is disposed in the first portion **690** of the proximal portion **624** of the cavity mixing chamber **618**.

(52) As the dynamic member **518** moves from the first configuration to the second configuration, the internal pressure in the first portion **690** of the proximal portion **624** of the cavity mixing chamber **618** decreases to a level less than the environment external to the cartridge housing **514**. This causes the sealing membrane **550** to move from the second configuration to the first configuration. This, in turn, causes the internal pressure in the second portion **692** of the proximal portion **624** of the cavity mixing chamber **618** to decrease to a level less than the pressure of the environment external to the cartridge housing **514** causing tattoo ink to be drawn into the cavity **536** through one or more passageways **752** of the plurality of passageways **750** and/or the opening **522** defined on the cartridge distal end **520**.

(53) As the dynamic member **518** moves from the second configuration to the first configuration, the internal pressure in the first portion **690** of the proximal portion **624** of the cavity mixing chamber **618** increases to a level greater than the pressure of the environment external to the cartridge housing **514**. This causes the sealing membrane **550** to move from the first configuration to the second configuration. This, in turn, causes the internal pressure in the second portion **692** of the proximal portion **624** of the cavity mixing chamber **618** to increase to a level greater than the pressure of the environment external to the cartridge housing **514** causing tattoo ink to be expelled out of the cavity **536** through one or more passageways **752** of the plurality of passageways **750** and/or the opening **522** defined on the cartridge distal end **520**.

(54) While cartridge **512** has been described as including a single sealing membrane **550**, any suitable number of sealing membranes can be used and a skilled artisan will be able to select an appropriate number based on various considerations. Example numbers of sealing membranes considered suitable include one, more than one, a plurality, two, three, or any other number of sealing membranes considered suitable for a particular embodiment.

(55) FIG. **8** illustrates another example embodiment of a tattoo ink mixing apparatus **810**. Tattoo ink mixing apparatus **810** is similar to the tattoo ink mixing apparatus **10** illustrated in FIG. **1** and described above, except as detailed below. Reference numbers in FIG. **8** refer to the same structural element or feature referenced by the same numbers in FIG. **1**, offset by **800**. Thus, the tattoo ink mixing apparatus **810** comprises a cartridge **812** having a lengthwise axis **813**, a cartridge housing **814**, a plunger **816**, a dynamic member **818**, and a cartridge distal end **820** defining an opening **822**.

(56) In this embodiment, the cartridge **812** further defines a mixing tip **1000**. The mixing tip **1000** has a proximal end **1004**, a distal end **1006**, and an elongate body **1006** extending from the

proximal end **1004** to the distal end **1006**. The elongate body **1006** has a diameter **1010** and a length **1008** measured along the lengthwise axis **813**. The distal end **1006** defines a structural feature **1014** that has a diameter **1016**. The diameter **1016** of the structural feature **1014** is greater than the diameter **1010** of the elongate body **1006** of the mixing tip **1000**. Alternatively, the structural feature **1014** can have a diameter **1016** that is less than, equal to, or substantially equal to the diameter **1010** of the elongate body **1006**.

(57) The mixing tip **1000** can have a distal end **1006** defining any structural feature **1014** considered suitable for a particular embodiment and skilled artisans will be able to select a suitable structural feature based on various considerations. Example structural features that are considered suitable include a blade.

(58) The elongate body **1006** has a diameter **1010** that is less than the diameter **950** of the opening **822** defined on the cartridge distal end **820**. In the illustrated embodiment, the proximal end **1004** is inserted through the opening **822** and is disposed in the cavity mixing chamber **918** such that a portion **1012** of the elongate body **1006** of the mixing tip **1000** is disposed in the cavity mixing chamber **918**. The elongate body **1006** defines a length **1008** that extends from the proximal end **1002** of the mixing tip **1000** to the distal end **1004** of the mixing tip **1000**.

(59) In this embodiment, the cavity **836** defines a cavity mixing chamber **918** and a cavity proximal chamber **920**. The cavity mixing chamber **918** extends from a first location **922** between the housing proximal end **824** and the housing distal end **826** to the opening **822** defined on the cartridge distal end **820**. The mixing chamber **918** defines a length **924** that is less than the length **1008** of the elongate body **1006** of the mixing tip **1000**. The cavity mixing chamber **918** is adapted to receive the proximal end **1002** of the mixing tip **1000** and a portion **1012** of the elongate body **1006** of the mixing tip **1000** such that the proximal end **1002** and the portion **1012** of the elongate member **1006** are disposed in the cavity mixing chamber **918**.

(60) The cavity proximal chamber **920** extends from the opening **830** defined on the housing proximal end **824** to the first location **922**. The cavity proximal chamber **920** has a first diameter **921** and is adapted to receive the second portion **868** of the plunger **816**.

(61) In this embodiment, the center portion **890** of the dynamic member **818** comprises a gearing **1020** that converts the linear motion of the plunger **816** into rotational motion for the mixing tip **1000**, i.e. the gearing **1020** transforms axial motion into radial motion. In this embodiment, the gearing **1020** has a first side **904** and a second side **906**. The first side **904** is adapted to receive the distal end **862** of the plunger **816**. The second side **906** is adapted to receive the proximal end **1002** of the mixing tip **1000**.

(62) The gearing **1020** can comprise any machine or part that transforms axial motion of the plunger **816** along the lengthwise axis **813** into radial motion for the mixing tip **1000** along the lengthwise axis **813**. The radial motion causes the proximal end **1002**, the elongate body **1006**, and the distal end **1006** with the structural feature **1014** to rotate about the lengthwise axis **813**. The direction of rotation depends on the type of gearing used and skilled artisans will be able to select an appropriate gearing to define a direction of rotation based on various considerations. Example directions of rotations that are considered suitable include clockwise, counter-clockwise, and both clockwise and counter-clockwise. For example, the gearing can be adapted to transform the proximal and distal linear motion of the plunger **816** to clockwise rotation, counter-clockwise rotation, and/or both clockwise and counter-clockwise rotation.

(63) Any gearing that transforms axial motion into radial motion can be used and skilled artisans will be able to select an appropriate gearing based on various considerations. Example gearings that are considered suitable include barrel cam, crank, or any other gearing considered suitable for a particular embodiment.

(64) FIG. 9 illustrates an example tattoo machine system **1100** useful for the application of tattoos and for mixing tattoo ink. The tattoo machine system **1100** comprises the tattoo ink mixing apparatus **810** illustrated FIG. 8, a tattoo machine **1110**, and a cartridge grip **1112**. The tattoo

machine system **1100** can also optionally include an ink reservoir **1052**. The tattoo ink mixing apparatus **810** comprises all the elements illustrated in FIG. **8** and described above. Thus, the tattoo ink mixing apparatus **810** comprises a cartridge **812** having a lengthwise axis **813**, a cartridge housing **814**, a plunger **816**, a dynamic member **818**, and a cartridge distal end **820** defining an opening **822**.

(65) The tattoo machine system **1100** can include any suitable tattoo machine **1110** and skilled artisans will be able to select an appropriate tattoo machine based on various considerations.

(66) The tattoo machine system **1100** can include any suitable cartridge grip **1112** and skilled artisans will be able to select an appropriate cartridge grip based on various considerations.

(67) The tattoo machine system **1100** can optionally include any ink reservoir **1052** and skilled artisans will be able to select an appropriate ink reservoir based on various considerations. The inclusion of an ink reservoir **1052** is completely optional and skilled artisans will be able to select whether or not to include an ink reservoir **1052** in a tattoo machine system **1100**.

(68) While tattoo machine system **1100** has been described as comprising the tattoo ink mixing apparatus **810** illustrated in FIG. **8**, the tattoo machine system **1100** can comprise any tattoo ink mixing apparatus. Skilled artisans will be able to select an appropriate tattoo ink mixing apparatus to be included in the tattoo machine system **1100** based on various considerations. For example, the tattoo machine system **1100** can comprise the tattoo ink mixing apparatus **10** illustrated in FIG. **1**, the tattoo ink mixing apparatus **110** illustrated in FIGS. **2**, **3**, and **4**, the tattoo ink mixing apparatus **510** illustrated in FIGS. **5**, **6**, and **7**, or any other tattoo ink mixing apparatus considered suitable for a particular embodiment.

(69) Those with ordinary skill in the art will appreciate that various modifications and alternatives for the described and illustrated embodiments can be developed in light of the overall teachings of the disclosure. Accordingly, the particular arrangements disclosed are intended to be illustrative only and not limiting as to the scope of the invention, which is to be given the full breadth of the appended claims and any and all equivalents thereof.

Claims

1. A tattoo ink mixing apparatus having a lengthwise axis and comprising: a housing having a housing proximal end, a housing distal end, a housing body extending from the housing proximal end to the housing distal end, the housing proximal end defining a housing proximal end opening, the housing distal end defining a housing distal end opening, the housing body defining a cavity extending from the housing proximal end opening to the housing distal end opening, the cavity defining a cavity proximal chamber and a cavity mixing chamber, the cavity proximal chamber extending from the housing proximal end opening to a first location between the housing proximal end and the housing distal end, the cavity mixing chamber extending from the first location to the housing distal end opening; a plunger having a plunger proximal end, a plunger distal end, and a plunger elongate body extending from the plunger proximal end to the plunger distal end, the plunger distal end disposed through the housing proximal end opening and in the cavity, the plunger adapted to move between a plunger first position in which the plunger distal end is disposed a first distance from the housing distal end and a second position in which the plunger distal end is disposed a second distance from the housing distal end, the first distance being greater than the second distance as measured along the lengthwise axis; a mixing tip having a mixing tip proximal end, a mixing tip distal end, and a mixing tip elongate body extending from the mixing tip proximal end to the mixing tip distal end, the mixing tip elongate body having a mixing tip elongate body diameter, the mixing tip proximal end disposed through the housing distal end opening and in the cavity, the mixing tip distal end defining a structural feature, the structural feature having a structural feature diameter that is greater than the mixing tip elongate body diameter; and a gearing adapted to convert linear motion of the plunger into rotational motion of

the mixing tip.

2. The tattoo ink mixing apparatus of claim 1, wherein the structural feature is a blade.

3. The tattoo ink mixing apparatus of claim 1, wherein a portion of the structural feature extends along a helical path.

4. The tattoo ink mixing apparatus of claim 1, wherein the gearing is a crank.

5. The tattoo ink mixing apparatus of claim 1, wherein the gearing is a barrel cam.

6. The tattoo ink mixing apparatus of claim 1, wherein the plunger has a diameter that is less than a first diameter of a housing proximal portion, the housing proximal portion extending from the housing proximal end to a housing intermediate portion, the housing intermediate portion extending from the housing proximal portion to a housing distal portion, the housing distal portion extending from the housing intermediate portion to the housing distal end.

7. The tattoo ink mixing apparatus of claim 1, wherein a plunger center point is disposed adjacent the housing proximal end opening when in a plunger starting configuration.

8. The tattoo ink mixing apparatus of claim 1, wherein a plunger center point is disposed proximal to the housing proximal end opening when in a plunger starting configuration.

9. The tattoo ink mixing apparatus of claim 1, wherein the plunger has a diameter that is substantially the same as a first diameter of a housing proximal portion, the housing proximal portion extending from the housing proximal end to a housing intermediate portion, the housing intermediate portion extending from the housing proximal portion to a housing distal portion, the housing distal portion extending from the housing intermediate portion to the housing distal end.

10. The tattoo ink mixing apparatus of claim 1, wherein the plunger has a diameter that is less than a first diameter of a housing proximal portion, the housing proximal portion extending from the housing proximal end to a housing intermediate portion, the housing intermediate portion extending from the housing proximal portion to a housing distal portion, the housing distal portion extending from the housing intermediate portion to the housing distal end.

11. The tattoo ink mixing apparatus claim 1, wherein the mixing tip elongate body diameter is less than the housing distal end opening.

12. The tattoo ink mixing apparatus of claim 1, wherein the housing has a housing proximal portion, a housing intermediate portion, and a housing distal portion, the housing proximal portion extending from the housing proximal end to the housing intermediate portion, the housing intermediate portion extending from the housing proximal portion to the housing distal portion, the housing distal portion extending from the housing intermediate portion to the housing distal end, the housing intermediate portion having a diameter that is greater than the diameter of the housing proximal portion.

13. The tattoo ink mixing apparatus of claim 1, wherein the housing has a housing proximal portion, a housing intermediate portion, and a housing distal portion, the housing proximal portion extending from the housing proximal end to the housing intermediate portion, the housing intermediate portion extending from the housing proximal portion to the housing distal portion, the housing distal portion extending from the housing intermediate portion to the housing distal end the housing intermediate portion and the housing distal portion having substantially the same diameter.

14. The tattoo ink mixing apparatus of claim 1, wherein the gearing has a first side and a second side.

15. The tattoo ink mixing apparatus of claim 14, wherein the first side is adapted to receive the plunger distal end.

16. The tattoo ink mixing apparatus of claim 14, wherein the second side is adapted to receive the mixing tip proximal end.

17. The tattoo ink mixing apparatus of claim 14, wherein the first side is adapted to receive the plunger distal end; and the second side is adapted to receive the mixing tip proximal end.

18. The tattoo ink mixing apparatus of claim 1, wherein the cavity proximal chamber having a cavity proximal chamber length along the lengthwise axis, the cavity mixing chamber having a

cavity mixing chamber length along the lengthwise axis, the cavity mixing chamber length is greater than the cavity proximal chamber length along the lengthwise axis.

19. A tattoo ink mixing apparatus having a lengthwise axis and comprising: a housing having a housing proximal end, a housing distal end, a housing body extending from the housing proximal end to the housing distal end, the housing proximal end defining a housing proximal end opening, the housing distal end defining a housing distal end opening, the housing body defining a cavity extending from the housing proximal end opening to the housing distal end opening, the cavity defining a cavity proximal chamber and a cavity mixing chamber, the cavity proximal chamber extending from the housing proximal end opening to a first location between the housing proximal end and the housing distal end, the cavity mixing chamber extending from the first location to the housing distal end opening; a plunger having a plunger proximal end, a plunger distal end, and a plunger elongate body extending from the plunger proximal end to the plunger distal end, the plunger distal end disposed through the housing proximal end opening and in the cavity, the plunger adapted to move between a plunger first position in which the plunger distal end is disposed a first distance from the housing distal end and a second position in which the plunger distal end is disposed a second distance from the housing distal end, the first distance being greater than the second distance as measured along the lengthwise axis; a mixing tip having a mixing tip proximal end, a mixing tip distal end, and a mixing tip elongate body extending from the mixing tip proximal end to the mixing tip distal end, the mixing tip elongate body having a mixing tip elongate body diameter, the mixing tip proximal end disposed through the housing distal end opening and in the cavity, the mixing tip distal end defining a structural feature, the structural feature having a structural feature diameter that is less than the mixing tip elongate body diameter; and a gearing adapted to convert linear motion of the plunger into rotational motion of the mixing tip.

20. A tattoo ink mixing apparatus having a lengthwise axis and comprising: a housing having a housing proximal end, a housing distal end, a housing body extending from the housing proximal end to the housing distal end, the housing proximal end defining a housing proximal end opening, the housing distal end defining a housing distal end opening, the housing body defining a cavity extending from the housing proximal end opening to the housing distal end opening, the cavity defining a cavity proximal chamber and a cavity mixing chamber, the cavity proximal chamber extending from the housing proximal end opening to a first location between the housing proximal end and the housing distal end, the cavity mixing chamber extending from the first location to the housing distal end opening; a plunger having a plunger proximal end, a plunger distal end, and a plunger elongate body extending from the plunger proximal end to the plunger distal end, the plunger distal end disposed through the housing proximal end opening and in the cavity, the plunger adapted to move between a plunger first position in which the plunger distal end is disposed a first distance from the housing distal end and a second position in which the plunger distal end is disposed a second distance from the housing distal end, the first distance being greater than the second distance as measured along the lengthwise axis; a mixing tip having a mixing tip proximal end, a mixing tip distal end, and a mixing tip elongate body extending from the mixing tip proximal end to the mixing tip distal end, the mixing tip elongate body having a mixing tip elongate body diameter, the mixing tip proximal end disposed through the housing distal end opening and in the cavity, the mixing tip distal end defining a structural feature, the structural feature having a structural feature diameter that is equal to the mixing tip elongate body diameter; and a gearing adapted to convert linear motion of the plunger into rotational motion of the mixing tip.
